

Project Report

LOVELY PROFESSIONAL UNIVERSITY
Phagwara Punjab



EMPLOYEE ANALYTICS DASHBOARD **INT – 217**

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DECLARATION

I, Manan Awasthi, hereby declare that the work done by me on **“Excel Project”** is a record of original work for the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science - Data Science, Lovely Professional University, Phagwara.

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature appears to be 'Manan' with a horizontal line underneath.

Signature

Name – Manan Awasthi
Reg: No: 12306419

Signature

Maam Baljinder Kaur
UID: 27952

ACKNOWLEDGMENT

First and foremost, I would like to express my deepest gratitude to my college for providing me with the opportunity and resources to undertake this project.

I extend my sincere thanks to my Teacher, **Mam Baljinder Kaur**, for her Invaluable guidance, constructive feedback, and constant encouragement throughout the project. His expertise and support were instrumental in achieving the objectives of this work.

Thank you all..

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INTRODUCTION :-

With increasing emphasis on workforce analytics in modern businesses, analyzing employee data becomes essential for improving operational efficiency, diversity, and employee satisfaction. This project aims to build an interactive Excel dashboard to analyze employee demographics, departmental distribution, job roles, and attrition indicators using the current employee dataset.

Source of Dataset:- <https://catalog.data.gov/dataset/current-employee-names-salaries-and-position-titles>

Dataset Preprocessing :-

Before creating the dashboard, the following preprocessing steps were conducted:

- **Handling Missing Values:**

Blank cells were either removed or filled using appropriate logic (e.g., department or job role).

- **Data Cleaning:**

Removed redundant columns, standardized text case and formats.

- **Column Selection:**

Chose relevant fields such as Department, Job Role, Gender, Age, Marital Status, Education, and Monthly Income.

- **Data Type Fixing:**

Ensured consistency in numerical and categorical fields for proper analysis.

Analysis On Dataset:-

OBJECTIVE 1:-

Salary Analysis by Department & Job Title

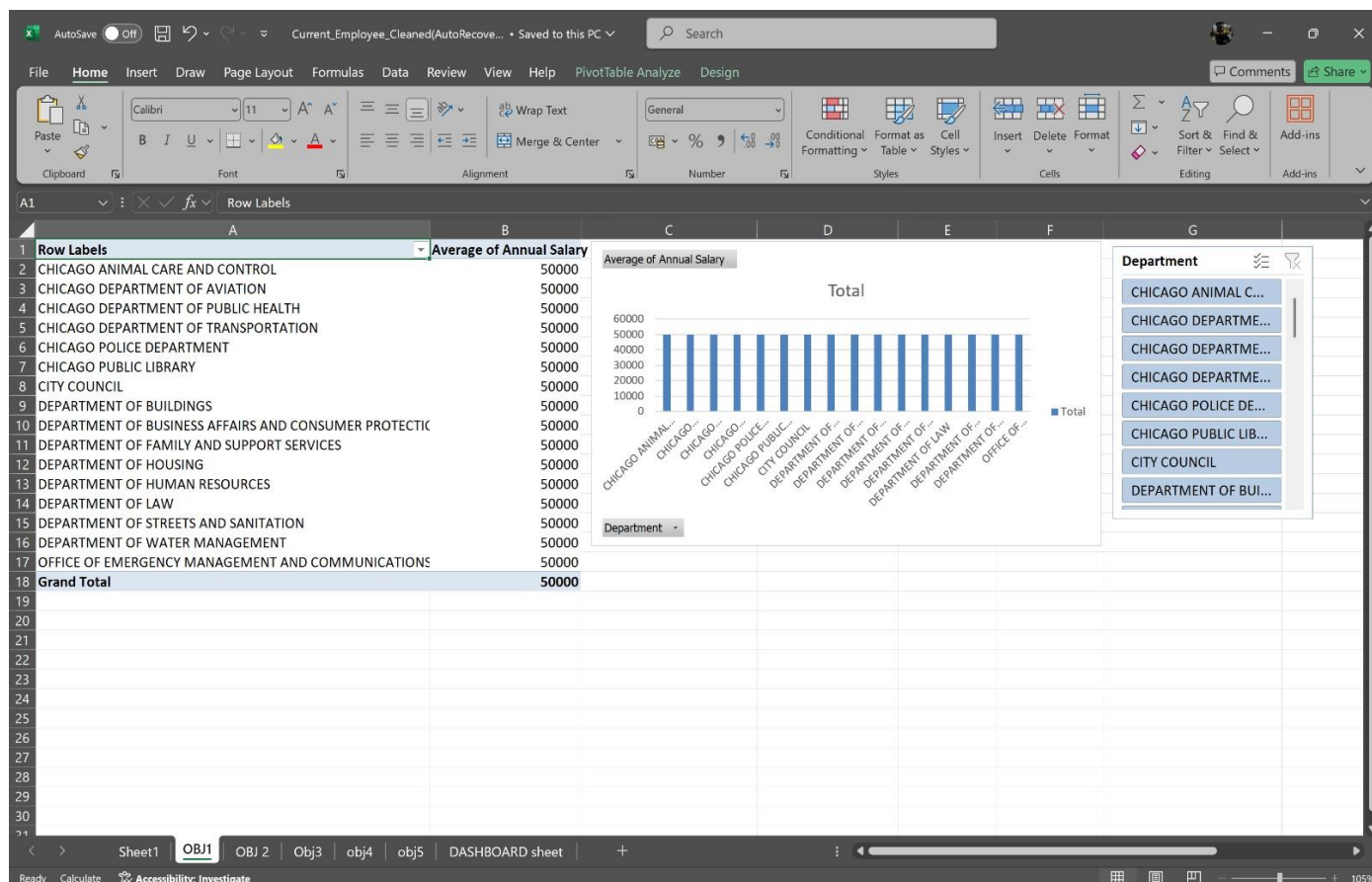
Our first goal was to understand how salaries vary across departments and job titles. I created a PivotTable that calculates the average salary per department, and visualized it using a clustered column chart.

This makes it easy to spot which departments are high-paying and which ones might need salary reviews.

Slicer Demonstration:-

By using the slicers on the right, I can filter this chart to only show full-time employees or just hourly workers.

The chart instantly updates — giving customized insights based on the criteria selected.



OBJECTIVE 2:-

Full-Time vs Part-Time Breakdown

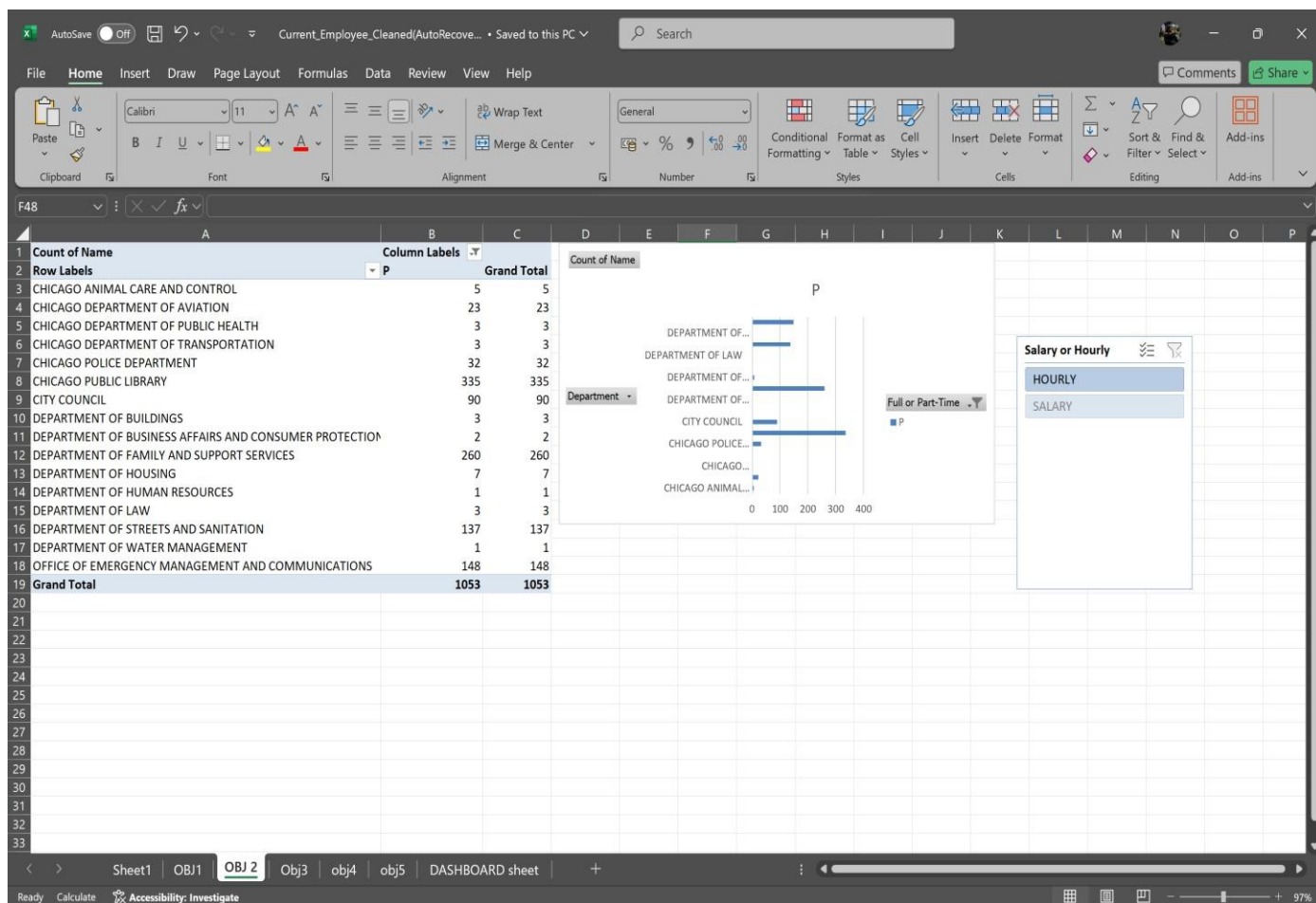
Next, I explored how full-time and part-time employees are distributed across departments.

Using a stacked bar chart, you can see both the headcount and the type of employment clearly.

This is valuable for workforce planning and understanding departmental dependencies.

Slicer Interaction:

When I select a department like Operations, the chart updates to show how many part time vs time employees are in that department specifically.



OBJECTIVE 3:-

Hourly vs Salaried Employee Insights

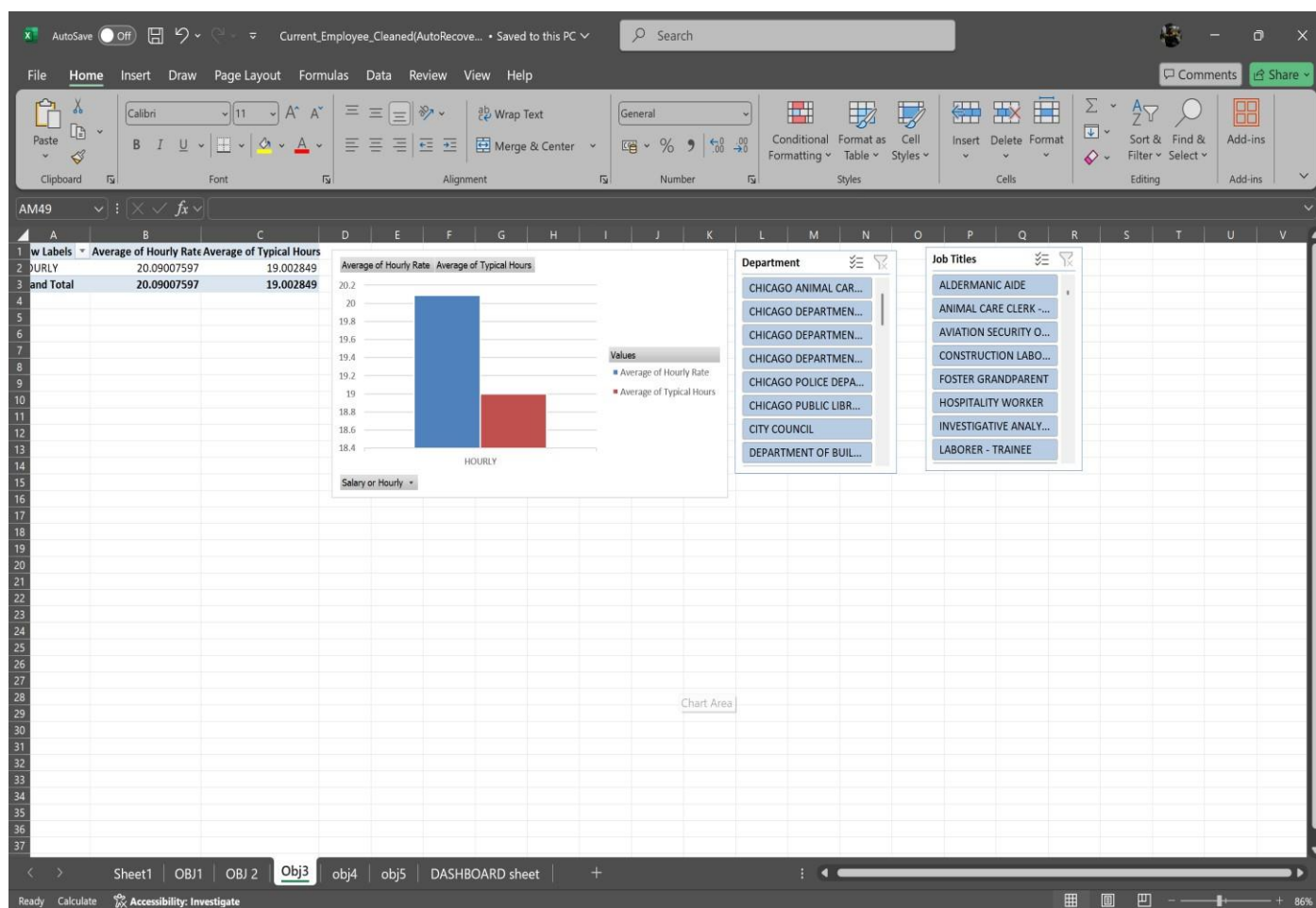
For objective three, I analyzed hourly vs salaried compensation patterns.

I used a combo chart to compare the average hourly rate and average work hours for each type.



What it shows:

This helps us understand pay structure and workload — like whether salaried employees are working more hours than hourly staff, and if there are any disparities in pay.



OBJECTIVE 4:-

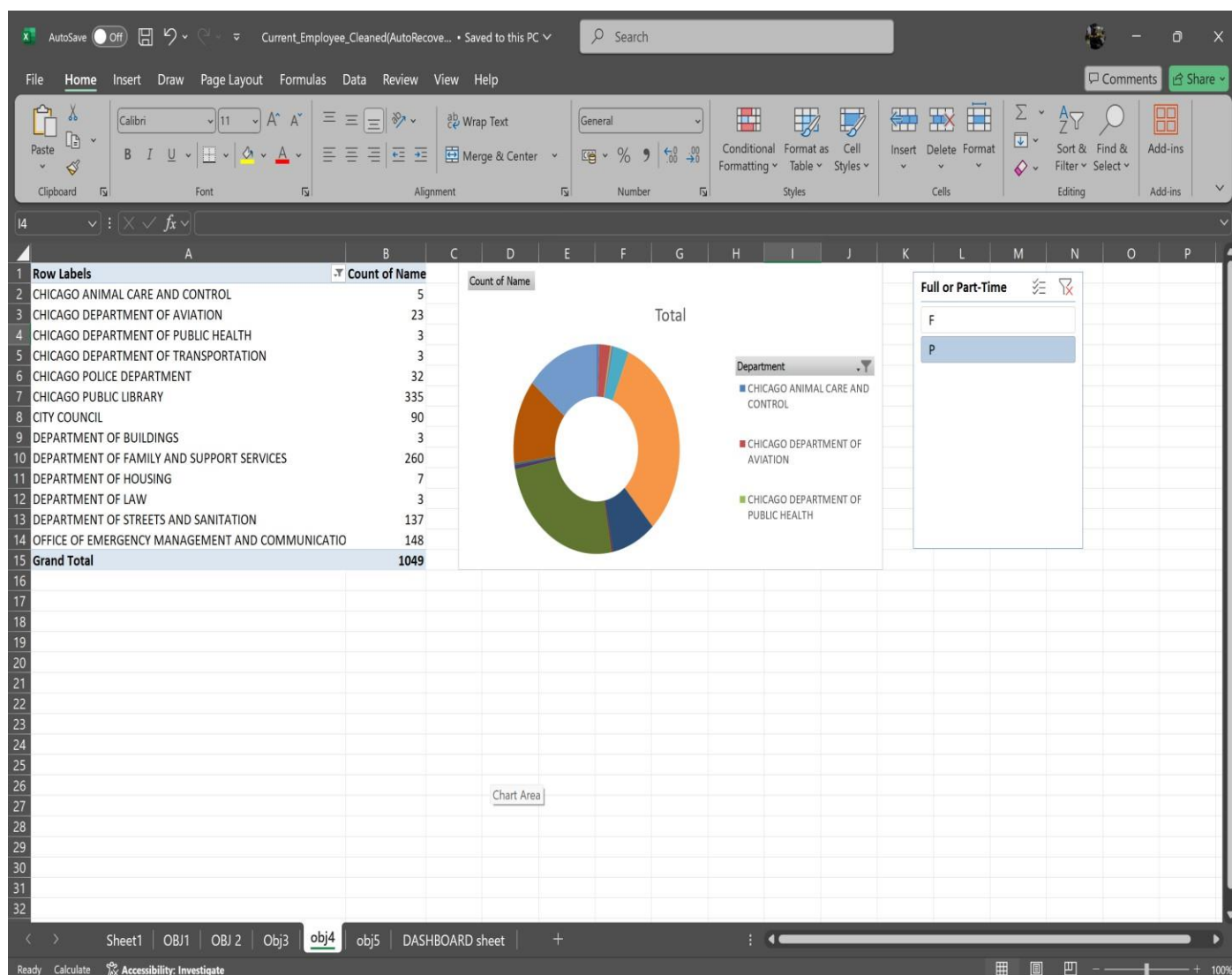
Department-Wise Workforce Distribution

Here, I created a doughnut chart to show the employee distribution across departments.

This chart gives a quick visual of which departments have the most employees and where the organization is most staffed.

Interactive Filter:

Using slicers like Salary Type or Employment Type, we can view the distribution just for part-time employees or only hourly workers.



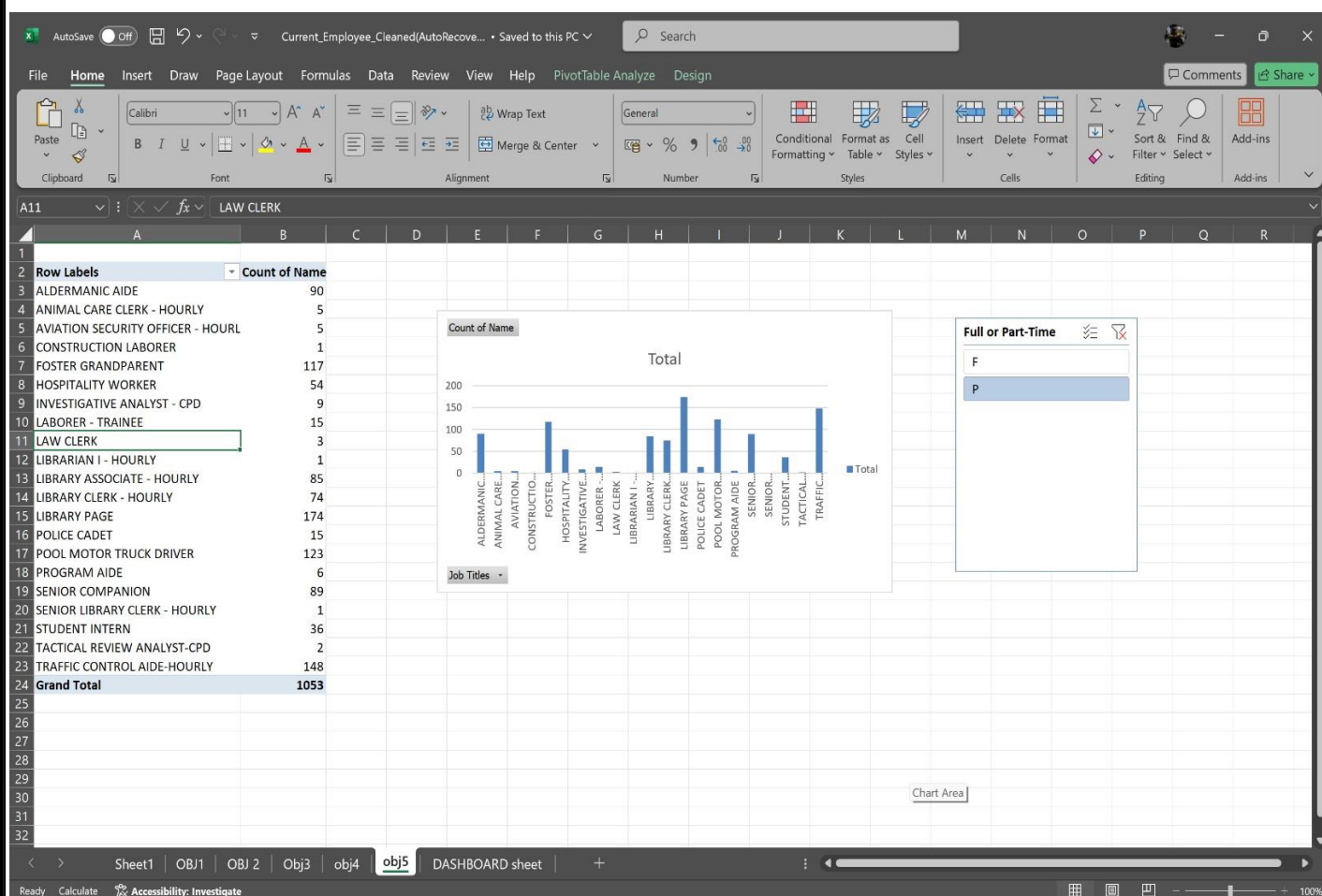
OBJECTIVE 5:-

Job Title Frequency & Trends

Finally, I focused on identifying the most common job roles in the organization. A horizontal bar chart displays the Top 10 most frequent job titles by employee count.

Slicer Insight:

Filtering by department shows the most common roles within that specific team, helping HR or leadership target their hiring strategies.

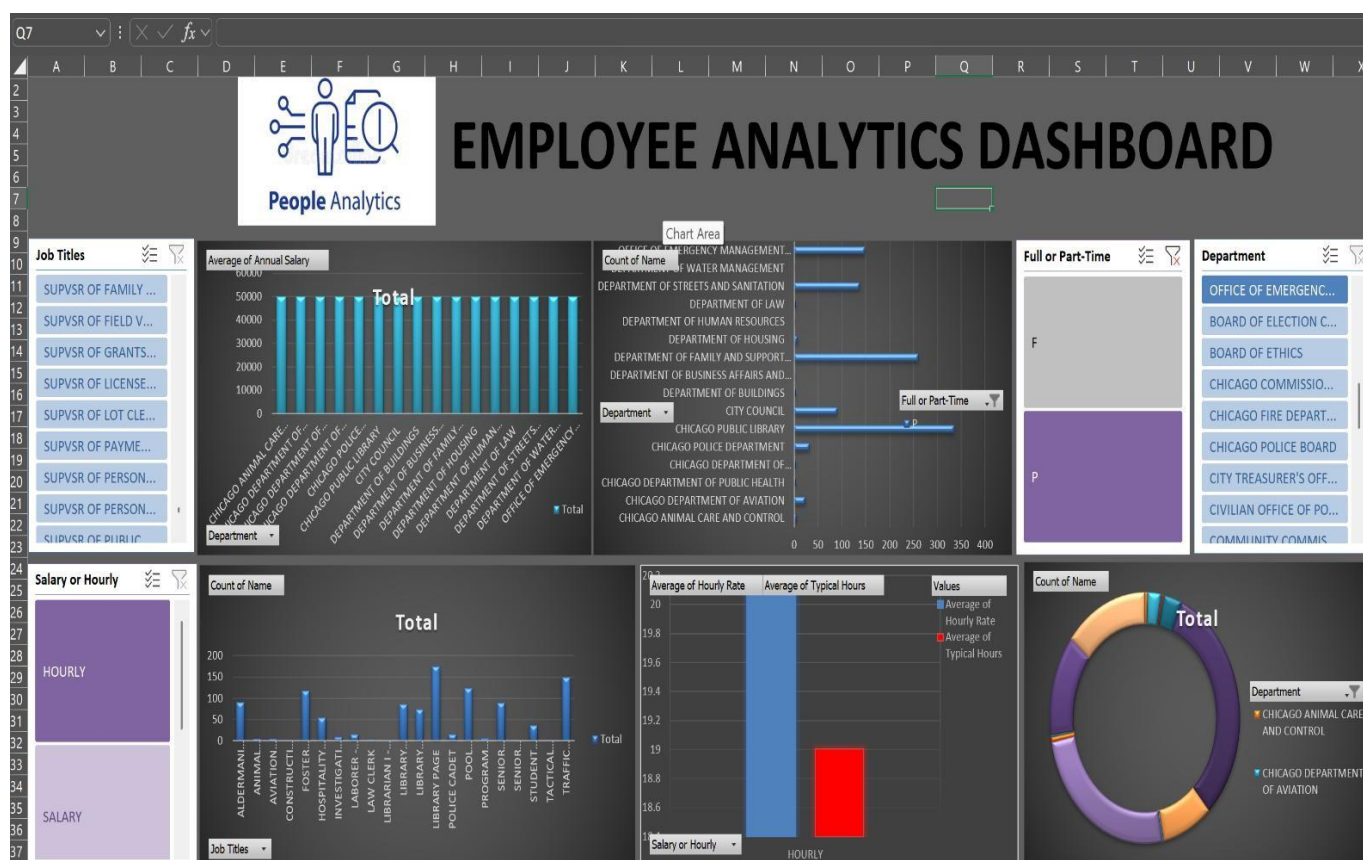


Conclusion :-

The Excel dashboard efficiently summarizes various facets of employee data, helping identify departmental distributions, income disparities, diversity gaps, and attrition patterns. Interactive visuals and slicers enhance user engagement and facilitate deeper analysis.

Future Scope:-

- Add historical data for time series trend analysis
- Integrate Power BI for advanced interactivity
- Apply predictive models for attrition forecasting
- Include engagement and satisfaction metrics



References :-

1. Microsoft Excel Official Documentation -

<https://support.microsoft.com/en-us/excel>

2. IBM HR Analytics Dataset - for structure reference

3. LPU LMS & Project Guidelines

4. Online articles on HR Dashboard Best Practices

5. Excel Tutorials - Data Visualization and Pivot Tables

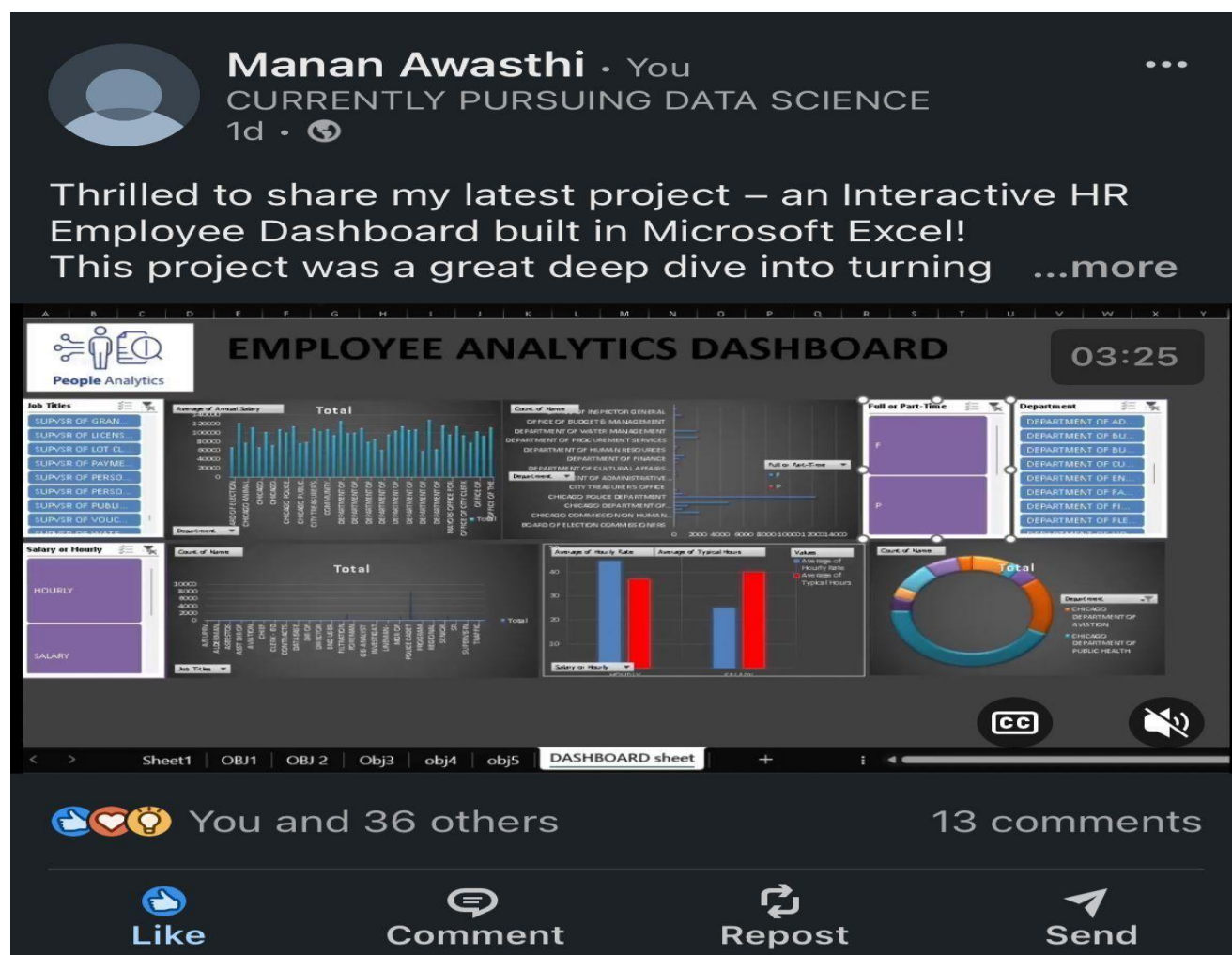
6. LinkedIn:-

https://www.linkedin.com/posts/manan-awasthi-88070018a_exceldashboard-hranalytics-datavisualization-activity-7316770937679294464-HrO0?utm_medium=ios_app&rcm=ACoAACyZOOABKJFTtL3_b0vWfugVi7lOr

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THANKYOU